

2023

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## Question 1

### 0.0.1 (a)

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$$\Xi = \prod_i \left(1 - e^{-\beta(\epsilon_i - \mu)}\right)^{-1} \quad (1)$$

$$\ln \Xi = - \sum_i \ln 1 - e^{-\beta(\epsilon_i - \mu)} \quad (2)$$

$$(3)$$

From lecture notes we have

$$\omega(\epsilon) = \delta(\epsilon) + \sum_{i \neq 0} \delta(\epsilon - \epsilon_i) \quad (4)$$

We assume energy differences are small

$$\omega(\epsilon) = \delta(\epsilon) + \deg \int_{-\infty}^{\infty} \delta(\epsilon - \epsilon_i) di^2 \quad (5)$$

We know that  $j = \frac{2\pi i}{L}$  and  $p = \hbar k \implies i = \frac{pL}{\hbar}$

$$= \delta(\epsilon) + \deg \frac{L^2}{h^2} \int_{-\infty}^{\infty} \delta\left(\epsilon - \frac{p^2}{2m}\right) d^2 p \quad (6)$$

$$(7)$$

Using polar

$$= \delta(\epsilon) + \deg \frac{2\pi L^2}{h^2} \int_0^{\infty} d|p| \delta\left(\epsilon - \frac{|p|^2}{2m}\right) p \quad (8)$$

$$= \delta(\epsilon) + \frac{2\pi L^2}{h^2} \deg \int_0^{\infty} \frac{du}{\sqrt{u}} \left(\frac{\sqrt{2m}}{2}\right) \sqrt{2mu} \quad (9)$$

$$\omega(\epsilon) = \delta(\epsilon) + \frac{2\pi L^2 m}{h^2} \deg \quad (10)$$